Open

exp9.l
~/Music/Exp86

exp9.l

```
%{
#include "y.tab.h"
#include <stdlib.h>
%}

%%

"for"       { return FOR; }

"("         { return LPAREN; }
")"         { return RPAREN; }
";"         { return SEMICOLON; }

"="         { return ASSIGN; }

"<"         { return LT; }
">"         { return GT; }

"++"        { return INC; }
"--"        { return DEC; }

[0-9]+      { yylval = atoi(yytext); return NUMBER; }

[a-zA-Z][a-zA-Z0-9]* {
                return ID;
            }

[ \t\n]+    ;

.           { return yytext[0]; }

%%

int yywrap()
{
    return 1;
}
```

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Batch: B1
Date of Execution: 06/10/26

sit@sit-OptiPlex-3280-AIO: ~/Music/Exp86

```
sit@sit-OptiPlex-3280-AIO:~/Music/Exp86$ yacc -d exp9.y
sit@sit-OptiPlex-3280-AIO:~/Music/Exp86$ flex exp9.l
sit@sit-OptiPlex-3280-AIO:~/Music/Exp86$ gcc y.tab.c lex.yy.c -ll
sit@sit-OptiPlex-3280-AIO:~/Music/Exp86$ ./a.out
Enter a for loop:
for(i = 10; i > n; i--)
Valid FOR loop statement
for(i = 0; i < n; i--)
Invalid FOR loop statement
sit@sit-OptiPlex-3280-AIO:~/Music/Exp86$
```

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Open

exp9.y

~/Music/Exp86

exp9.y

```
%{
#include <stdio.h>
#include <stdlib.h>

int yylex();
void yyerror(const char *s);
%}

%token FOR
%token LPAREN RPAREN SEMICOLON
%token ASSIGN LT GT
%token INC DEC
%token ID NUMBER

%%

program:
        for_statement
        {
            printf("Valid FOR loop statement\n");
        }
        ;

for_statement:
        FOR LPAREN initialization SEMICOLON condition SEMICOLON increment RPAREN
        ;

initialization:
        ID ASSIGN NUMBER
        ;

condition:
        ID LT NUMBER
        | ID GT NUMBER
        | ID LT ID
        | ID GT ID
        ;

increment:
        ID INC
        | ID DEC
        ;
```

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